UNIT 5 – INTRODUCTION TO MACHINE LEARNING

What is Machine Learning?

Field of study that gives computers a capability to learn without being explicitly programmed.

Eg: online shopping

Machine will adapt the pattern based on data feed by user.

Machine learning (ML) is a subset of artificial intelligence (AI) and is dedicated to making computers and machines mimic human learning, acting on tasks autonomously, and enhancing their work and accuracy as they gain experience and are subjected to more information.

UC Berkeley separates the machine learning system of a machine learning algorithm into three primary components.

1. A Decision Process: For the most part, machine learning algorithms provide a prediction or classification. Based on some kind of input data, which may be labeled or unlabeled, your algorithm will provide an estimate regarding a pattern in the data.

2. An Error Function: An error function examines the model's prediction. Where there are known examples, an error function can compare in order to determine the model's accuracy.

3. A Process of Model Optimization: If the model can fit closer to the data points in the training set, then weights are modified to decrease the difference between the known example and the model estimate. The algorithm will continue this iterative "evaluate and optimize" cycle, adjusting weights independently until a stopping point of accuracy has been reached.

Traditional, or "non-deep," machine learning is more reliant on human effort to learn. Human professionals specify the feature set in order to learn the variations between data inputs, typically needing more structured data to learn

Deep neural networks and neural networks have been hailed for speeding up advancements in fields like computer vision, natural language processing (NLP), and speech recognition.

Machine learning methods :

Machine learning models fall into **three primary categories**.

1. **Supervised learning** -

Supervised learning, or supervised machine learning, is characterized by its utilization of labeled datasets to train algorithms to classify data or predict outcomes correctly. As input data is provided to the model, the model tweaks its weights until it has been suitably fitted. This happens as part of the cross validation exercise to make sure that the model does not overfit or underfit. Supervised learning assists organizations in addressing a range of real-world problems at scale, like spam classification in a folder independent of your inbox. Neural networks, Naïve Bayes, linear regression, logistic regression, random forest, and support vector machine (SVM) are some of the techniques applied in supervised learning.

2. Unsupervised learning

Unsupervised learning, or unsupervised machine learning, employs the use of machine learning algorithms to cluster and analyze unlabeled datasets (subsets referred to as clusters). Unsupervised learning identifies concealed patterns or groupings of data without human involvement.

Unsupervised learning's capacity to find similarities and differences between information make it best suited to exploratory data analysis, cross-selling campaigns, customer segmentation, and image and pattern recognition. It's also applied to decrease the amount of features in a model via the method of dimensionality reduction. Two popular methods for this are principal component analysis (PCA) and singular value decomposition (SVD). Other algorithms applied in unsupervised learning are neural networks, k-means clustering, and probabilistic clustering techniques.

3. Semi-supervised learning

Semi-supervised learning provides a pleasant middle ground between supervised and unsupervised learning. During the training process, it employs a small labeled set of data to inform classification and feature extraction from a large, unlabeled set of data. Semi-supervised learning can address the issue of having inadequate labeled data for a supervised learning algorithm. It's also beneficial if it's too expensive to label sufficient data.

4. **Reinforcement learning**

Reinforcement learning is a machine learning framework that closely resembles supervised learning, but the algorithm is not trained based on sample data. This framework learns as it progresses through trial and error. A series of positive outcomes will be rewarded to build the best recommendation or policy for a particular problem.

For a dive into the distinctions between these methods, see "Supervised vs. Unsupervised Learning: What's the Difference?"

Common machine learning algorithms

A number of machine learning algorithms are commonly used. These include:

- Neural networks
- Linear regression
- Logistic regression
- Clustering
- Decision trees
- Random forests

Common ML Workflow:

Data Collection – Gather relevant data.

Data Preprocessing – Clean, normalize, and format data.

Feature Selection/Engineering – Choose important variables.

Model Selection – Choose an algorithm (e.g., Decision Tree, SVM).

Training – Fit the model using training data.

Testing/Evaluation – Measure performance using unseen test data.

Prediction – Use the model for real-world decisions.

What is Supervised Learning?

Supervised learning is a **type of machine learning** where the model is trained on a **labeled dataset**, meaning each training example has both:

- Input (features): Also called **independent variables** (e.g., age, income)

- Output (target label): Also called **dependent variable** (e.g., house price, disease class)

The goal is to learn a **mapping function** from input variables **X** to an output variable **Y**, written as:
Y = f(X)

* X: Input features (vector or matrix)

* Y: Target output or label

How Supervised Learning Works

- **Input data** with known output labels is fed into the model.
- The model makes **predictions** based on patterns it learns.
- **Loss/Cost function** measures the difference between predicted and actual outputs.
- The model is **trained iteratively** to minimize the loss using optimization techniques like **Gradient Descent**.
- Once trained, the model can be used to predict outcomes on **new, unseen data**.

⍰ Supervised Learning Workflow

1. **Data Collection**
2. **Data Preprocessing**

    - Handle missing values
    - Normalize/standardize
    - Encode categorical data

3. **Splitting Data -** Training Set (e.g., 80%) , Testing Set (e.g., 20%)
4. **Model Selection**
5. **Training the Model**
6. **Model Evaluation**
7. **Model Deployment**

In **supervised learning**, the algorithm is trained using **labelled data**, which means the input data is already tagged with the correct output.

🔹 **Example:** Think of it like a student learning from a teacher. The teacher (dataset) provides questions (input) and answers (output), and the student (model) learns the pattern.

Once trained, it can predict the output for new input data.

✨ *Applications:*

- Spam email detection
- Medical diagnosis
- Stock price prediction
- Customer churn prediction

📂 *Supervised Learning has two main types:*

A. Classification:

- Predicts **discrete** categories or classes.
- E.g., Email is spam or not, fruit is apple or banana.
- **Algorithms:** SVM, KNN, Naive Bayes

B. Regression:

- Predicts **continuous** values.
- E.g., Price of a product, temperature, sales forecasting.
- **Algorithms:** Linear Regression, Neural Networks

💡 Use supervised learning when both input and output data are available during training.

What is Unsupervised Learning?

In **unsupervised learning**, the algorithm is given **unlabelled data** — meaning there are no outputs provided, and the machine has to learn the structure on its own.

🔹 It tries to **identify patterns or groupings** in the data.

🔹 Useful when we don't know the outcomes in advance.

✨ *Applications:*

- Market segmentation
- Recommendation systems
- Customer behavior analysis

🔹 The machine groups similar items based on features, such as shape or color, without being told what each item is.

Example: If given a mix of apples and mangoes, the machine will group similar fruits without knowing their names.

📌 *Visual from Figure 9.2:*

- Input: Unlabeled fruits
- Model: Finds patterns
- Output: Grouped fruits

💡 **U**se unsupervised learning when you only have input data and want to find hidden patterns.

Unsupervised learning involves several algorithms and techniques designed to find patterns and structures in unlabeled data. Some key unsupervised learning methods include:

Clustering, Association Rule Learning, Dimensionality Reduction, Restricted Boltzmann Machines (RBMs).

Clustering

Clustering is a type of unsupervised machine learning algorithm that groups similar data points or observations into clusters. The goal of clustering is to identify patterns or structures in the data that are not easily visible by other means.

Types of Clustering

1. Hierarchical Clustering: This type of clustering builds a hierarchy of clusters by merging or splitting existing clusters.

2. K-Means Clustering: This is a popular clustering algorithm that partitions the data into K clusters based on the mean distance of the features.

3. DBSCAN (Density-Based Spatial Clustering of Applications with Noise): This algorithm clusters data points based on density and proximity to each other.

4. K-Medoids Clustering: This is a variation of K-Means clustering that uses medoids (objects that are representative of their cluster) instead of centroids.

How Clustering Works

1. Data Preprocessing: The data is preprocessed to remove any noise or irrelevant information.

2. Distance Measurement: A distance metric is chosen to measure the similarity between data points.

3. Clustering Algorithm: A clustering algorithm is applied to the data to identify clusters.

4. Cluster Evaluation: The quality of the clusters is evaluated using metrics such as silhouette score or Calinski-Harabasz index.

Applications of Clustering

1. Customer Segmentation: Clustering can be used to segment customers based on their behavior, demographics, or preferences.

2. Image Segmentation: Clustering can be used to segment images into different regions based on pixel values.

3. Gene Expression Analysis: Clustering can be used to identify patterns in gene expression data.

4. Anomaly Detection: Clustering can be used to identify outliers or anomalies in the data.

Key Concepts

1. Cluster: A group of similar data points or observations.

2. Centroid: The mean or average value of a cluster.

3. Distance Metric: A measure used to calculate the similarity between data points.

4. Dimensionality Reduction: Techniques used to reduce the number of features in the data.

Advantages and Disadvantages of Clustering

Advantages

1. Unsupervised Learning: Clustering is an unsupervised learning technique, meaning it doesn't require labeled data.

2. Pattern Discovery: Clustering can help discover patterns or structures in the data.

3. Data Reduction: Clustering can help reduce the complexity of large datasets.

Disadvantages:

1. Choosing the Right Algorithm: Choosing the right clustering algorithm can be challenging.

2. Interpreting Results: Interpreting the results of clustering can be difficult.

3. Sensitivity to Parameters: Clustering algorithms can be sensitive to parameter settings.

✅ Concluded:

Type	Training Data	Goal	Examples
Supervised Learning	Labeled (X, Y)	Predict known outputs	Spam detection, price prediction
Unsupervised Learning	Unlabeled (Only X)	Discover patterns or clusters	Customer segmentation, grouping

What is Reinforcement Learning?

Reinforcement Learning (RL) is a subfield of machine learning that involves training an agent to make decisions in an environment to maximize a reward. The agent learns through trial and error by interacting with the environment and receiving feedback in the form of rewards or penalties.

Key Concepts

1. Agent: The decision-making entity that interacts with the environment.

2. Environment: The external world that the agent interacts with.

3. Actions: The decisions made by the agent in the environment.

4. States: The current situation or status of the environment.

5. Rewards: The feedback received by the agent for its actions, which can be positive (reward) or negative (penalty).

6. Policy: The strategy used by the agent to select actions based on the current state.

How RL Works?

1. The agent observes the current state of the environment.

2. The agent selects an action based on its policy.

3. The agent takes the action and observes the next state and reward.

4. The agent updates its policy based on the reward received.

5. Steps 1-4 are repeated until the agent achieves its goal or a termination condition is met.

Types of RL

1. Episodic RL: The agent learns from a sequence of episodes, where each episode consists of a single interaction with the environment.

2. Continuous RL: The agent learns from a continuous stream of interactions with the environment.

Applications of RL

1. Game Playing: RL has been used to play complex games like Go, Poker, and Video Games.

2. Robotics: RL is used to control robots and optimize their performance in tasks like manipulation and navigation.

3. Recommendation Systems: RL can be used to personalize recommendations based on user behavior.

4. Autonomous Vehicles: RL is being explored for use in autonomous vehicles to improve decision-making.

Advantages of RL

1. Flexibility: RL can be applied to a wide range of problems.

2. Autonomy: RL enables agents to learn and adapt without human intervention.

3. Optimization: RL can optimize performance in complex environments.

Challenges of RL

1. Exploration-Exploitation Trade-off: The agent must balance exploring new actions and exploiting known actions.

2. High-Dimensional State and Action Spaces: RL can be challenging in environments with large state and action spaces.

3. Delayed Rewards: RL can struggle with delayed rewards, where the reward is received after a long sequence of actions.

* Reinforcement Learning is a powerful technique for training agents to make decisions in complex environments. By understanding the key concepts, types, and applications of RL, students can gain a deeper appreciation for the field and its potential to solve real-world problems.

Introduction to Scikit-learn

Scikit-learn is a popular open-source machine learning library for Python that provides a wide range of algorithms for classification, regression, clustering, and other tasks. It is widely used in industry and academia for data analysis, predictive modeling, and data science.

Key Features of Scikit-learn

1. Simple and Consistent API: Scikit-learn has a simple and consistent API that makes it easy to use and integrate with other libraries.

2. Wide Range of Algorithms: Scikit-learn provides a wide range of algorithms for classification, regression, clustering, and other tasks.

3. High-Quality Implementation: Scikit-learn algorithms are implemented with high-quality and efficient code.

4. Interoperability: Scikit-learn integrates well with other popular data science libraries in Python, such as NumPy, Pandas, and Matplotlib.

Some Popular Algorithms in Scikit-learn

1. Supervised Learning:

- Linear Regression

- Logistic Regression

- Decision Trees

- Random Forests

- Support Vector Machines (SVMs)

2. Unsupervised Learning:

- K-Means Clustering

- Hierarchical Clustering

- Principal Component Analysis (PCA)

3. Model Selection and Evaluation:

- Cross-Validation

- Grid Search

- Metrics for classification and regression

Advantages of Using Scikit-learn

1. Easy to Use: Scikit-learn has a simple and intuitive API that makes it easy to use.

2. Flexible: Scikit-learn provides a wide range of algorithms and tools that can be used for various tasks.

3. High-Quality Implementation: Scikit-learn algorithms are implemented with high-quality and efficient code.

4. Large Community: Scikit-learn has a large and active community that contributes to its development and provides support.

Use Cases for Scikit-learn

1. Predictive Modeling: Scikit-learn can be used to build predictive models for classification and regression tasks.

2. Data Analysis: Scikit-learn can be used for data analysis, feature selection, and dimensionality reduction.

3. Clustering: Scikit-learn can be used for clustering and identifying patterns in data.

Getting Started with Scikit-learn

1. Install Scikit-learn: Install Scikit-learn using pip or conda.

2. Import Scikit-learn: Import Scikit-learn in your Python script or notebook.

3. Load Data: Load your data into a Pandas dataframe or NumPy array.

4. Choose an Algorithm: Choose a Scikit-learn algorithm that suits your task.

5. Train and Evaluate: Train and evaluate your model using Scikit-learn's tools and metrics